

# **AI-Powered Image Captioning Using a Pre-Trained Vision-Language Model**

**Noah Paprocki**

Oakland University

npaprocki2@oakland.edu

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## *Abstract*

This project explores the use of a pre-trained vision language model for automatic image captioning. Using the BLIP (Bootstrapping Language-Image Pretraining) model implemented in PyTorch, the system generates natural language descriptions for a set of input images. A small dataset of diverse images was used to evaluate the model's performance. The generated captions were assessed using both manual quantitative metrics (accuracy, detail, clarity) and qualitative analysis. Results show that the model performs well on simple images with clear subjects but tends to produce generic captions for more complex scenes, and can sometimes focus on only a small part of the entire image.

## **1 Introduction**

Recent advances in artificial intelligence, particularly in generative models and vision language models, have enabled machines to understand and describe visual content in natural language. One important application of this capability is image captioning, which combines computer vision and natural language processing.

Image captioning has practical applications in accessibility, image search, and content organization. For example, it can assist visually impaired users by describing images or automatically tag large image datasets.

The goal of this project is to build a system that generates captions for images using a modern pre-trained model and to evaluate its performance on a small dataset.

## **Related Work**

Image captioning has been widely studied in the field of artificial intelligence. Earlier approaches typically combined Convolutional Neural Networks (CNNs) for image feature extraction with Recurrent Neural Networks (RNNs) for text generation.

One method of image captioning is attention-based captioning. It is inspired by the human attention pattern and the way the human eye focuses on images (Ghandi et al., 2023). The training process consists of showing the model where to look when shown an image. This makes it more natural to humans and more similar to how we look and process images.

This project builds upon these advancements by applying a pre-trained BLIP model to a custom dataset without additional fine-tuning.

## **Data**

The dataset used in this project consists of 15 images collected from a publicly available source: Unsplash.

The images were selected to represent a variety of categories, including:

- Animals

- People
- Food
- Outdoor scenes
- Everyday objects

This diversity helps evaluate how well the model generalizes across different types of visual content.

No preprocessing beyond resizing and converting images to RGB format was required, as the model's processor handles input formatting.

## **Methods**

This project uses the BLIP image captioning model (Salesforce/blip-image-captioning-base) implemented using PyTorch and the Hugging Face Transformers library.

BLIP is a vision language model trained on large datasets of image-text pairs. It learns to map visual features to natural language descriptions, enabling it to generate captions for unseen images.

## **Implementation**

The system follows these steps:

1. Load the pre-trained BLIP processor and model
2. Upload and store images in a working directory
3. Convert each image into a format suitable for the model
4. Generate captions using the model's `generate()` function
5. Decode the output tokens into readable text

Beam search (`num_beams=5`) was used during generation to improve caption quality by generating multiple outputs and picking the best one.

## Experiments and Results

### Caption Generation

The model successfully generated captions for all input images. Example outputs include:

**Figure 1:**

a woman and two children in a wagon picking apples from a tree



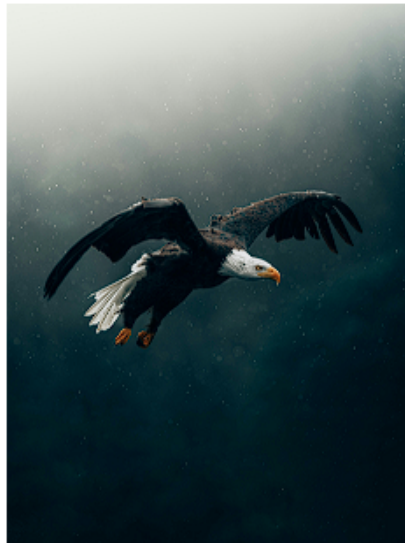
**Figure 2:**

a brown dog carrying a stick in the woods



**Figure 3:**

an eagle flying through the air with its wings spread



## **Quantitative Evaluation**

Each generated caption was manually evaluated using the following metrics:

- **Accuracy:** How well the caption matches the image
- **Detail:** How descriptive the caption is
- **Clarity:** How readable and grammatically correct the caption is

Scores were assigned on a scale from 1 to 5.

| Metric   | Average Score |
|----------|---------------|
| Accuracy | 4.53          |
| Detail   | 4.0           |
| Clarity  | 4.67          |

## Qualitative Analysis

The model performed well on images with clear and simple subjects, such as animals, food, and basic outdoor scenes. In these cases, captions were generally accurate and easy to understand.

However, for more complex images (especially ones with multiple subjects and more things going on), the model often generated generic descriptions or only focused on a small part and missed finer details or contextual relationships between objects. This suggests that while the model captures high-level features effectively, it has limitations in detailed scene understanding.

## Limitations

This project has several limitations:

- **Small dataset:** Only a limited number of images were used for evaluation
- **No fine-tuning:** The model was used as-is without adaptation to the dataset
- **Generic captions:** The model sometimes produces vague descriptions
- **Subjective evaluation:** Manual scoring introduces potential bias

## Conclusion

This project demonstrates that pre-trained vision-language models can effectively generate captions for images with minimal setup. The BLIP model was able to produce generally accurate and readable captions across a variety of images.

While the model performs well on simple images, it struggles with complex scenes and detailed descriptions. Future work could involve fine-tuning the model on a specific dataset, comparing multiple models, or improving evaluation methods.

## Future Work

Potential improvements include:

- Fine-tuning the model on a domain-specific dataset
- Comparing multiple captioning models
- Using automated evaluation metrics
- Expanding the dataset for better evaluation
- Building a user interface for real-time caption generation

## Citations And Acknowledgements

- BLIP model by Salesforce from Hugging Face
- Pytorch framework
- Images from Unsplash
- ChatGPT for ideas on image to text models

Ghandi, T., Pourreza, H., & Mahyar, H. (2023). *Deep learning approaches on image captioning: A review*. ACM Computing Surveys, 56(3). <https://doi.org/10.1145/3617592>